MediAI: Smart Patient Triage & Emergency Risk Prediction System

Master Project Specification (v2.0 – Production Baseline)

- **Document Status:** Baseline Single Source of Truth
- **Version:** 2.0
- **Target Environment:** Academic Final-Year Project, Clinical Decision-Support Prototyping, and Enterprise AI Development
- **Clinical Safety Disclaimer:** *MediAI is an AI-assisted clinical decision-support prototype. It must not independently diagnose conditions, prescribe medications, or replace certified clinical judgment.*

1. Project Identity & Governance

- **Purpose:** Establish a software platform for structured health-information capture, risk-oriented analysis, patient prioritization, and healthcare-professional decision support.
- **Vision:** A maintainable, secure, explainable, and extensible digital foundation for emergency triage and risk stratification.
- **Quality Characteristics:** Modular, secure, explainable, responsive, accessible, testable, observable, version-controlled, and documented.
- **Core Principles:**

– Separation of concerns and single responsibility.

– Explicit API, data, and component contracts.

– Deterministic safety over probabilistic machine learning predictions (hard red-flag overrides).

– Zero diagnostic or therapeutic authority (decision-support only).

– Human-in-the-loop oversight for all safety-critical triage decisions.

– Least privilege and zero plaintext Protected Health Information (PHI).

2. Project Scope & Operational Boundaries

In Scope

- Structured patient intake across web and mobile responsive interfaces.
- Automated evaluation across 12 clinical physiological risk categories.
- Deterministic physiological red-flag detection engine with immediate emergency escalation.
- Machine-learning risk scoring (0–100 scale) and 4-tier triage acuity stratification.
- Explainable AI (XAI) feature attribution reporting.

- Uncertainty and missing-data management.
- English and Tamil bilingual interface support.
- Voice-assisted symptom reporting.
- Attending physician triage review, queue management, and override audit ledger.
- Relational database schema, versioned migrations, and standardized REST API endpoints.
- Role-based access control (RBAC), field-level encryption, and tamper-evident audit logging.

Out of Scope (Explicit Non-Goals)

- Definitive disease diagnosis or automated medical report generation.
- Direct prescribing, dosage calculation, or dispensing of pharmaceutical drugs.
- Replacing certified emergency medical technicians (EMTs), triage nurses, or emergency physicians.
- Automated clinical overrides that silence critical physiological alarms.
- Direct automated integration with life-support telemetry or active infusion pumps.

3. Product Architecture & Layer Boundaries

The system strictly enforces unidirectional dependency flow: `Presentation` → `Application` → `Domain/Business` → `Data Access` → `Database`, with isolated cross-cutting adapters.

- **[Presentation Layer]**
  – Patient Self-Reporting Portal (Mobile/Desktop, Bilingual: EN / TA)
  – Nurse Triage Intake Terminal (Rapid input, vitals capture, voice intake)
  – Doctor Emergency Workstation (Priority queue, live alerts, override console)
- **[Application & Orchestration Layer]**
  – Authentication & RBAC Guard (JWT evaluation, session management)
  – Intake Flow Coordinator & Boundary Validation Pipeline
  – Alert Dispatcher & Notification Bus
- **[Clinical & Intelligence Layer]**
  – Safety Pre-Check Engine (Deterministic Hardcoded Red-Flag Filters)
  – ML Risk Prediction Service (Ensemble Acuity Classifier & Regressor)
  – Explainable AI Engine (SHAP / Feature Attribution Generator)
  – Clinical Decision Support (Vital sign delta analysis & contraindication alerts)
- **[Data Access & Integration Layer]**
  – Repository Layer (PostgreSQL with Row-Level Security)
  – De-identification Pipeline (Anonymized feature extraction for ML)
  – Audit Log Publisher (Tamper-evident append-only journal)

Layer Isolation Rules:

- Lower layers must never import or depend on higher layers.
- Presentation components must not contain business rules or direct persistence queries.
- Shared services must remain generic and never serve as dumping grounds for module-specific logic.
- External integrations (e.g., third-party speech recognition, SMS dispatchers) must sit behind isolated adapters.

4. Master Application Flow

User Access

Authentication & Role Resolution (Nurse, Doctor, Admin, or Anonymous Patient)

Application Shell & Navigation Guard

Patient Intake (Demographics, Chief Complaint, Symptom Checklist, Voice Input)

Physiological Vital Signs Capture

Safety Pre-Check Engine (Evaluates Tier-1 Red Flags)

– **[RED FLAG DETECTED]** → Bypass ML → Immediate Escalate to **CRITICAL (P1)**

– **[NO RED FLAG]** → Feature Preprocessing & Imputation Check

– ML Risk Model Inference (Calculates 0–100 Score & Acuity Band)

– Explainability Engine (Generates Top Risk Drivers via SHAP)

Persist Assessment, Vitals, and Audit Event

Real-Time Dispatch to Doctor Emergency Workstation Queue

Attending Physician Review & Sign-Off / Clinical Override (with mandatory rationale)

Final Disposition & Treatment Bay Allocation

5. The 12 Clinical Risk Domains & Red-Flag Safety Engine

A positive match on any Tier-1 Clinical Red Flag instantly forces a hardcoded triage status of Critical (Priority 1) and bypasses machine learning inference delays.

No.	Risk Domain	Required Inputs & Assessment Vectors	Tier-1 Red Flags (Hard Override to Critical)	Clinical Acuity Focus
1	General Infections & Sepsis	Core temp, HR, RR, BP, altered mentation, rigors, diaphoresis	qSOFA $\ge$ 2 (RR $\ge$ 22, SBP $\le$ 100, altered mentation), hypothermia ($<35^\circ\text{C}$)	Septic shock, systemic inflammatory response syndrome (SIRS)
2	Respiratory Warning Signs	SpO_2, RR, cough character, accessory muscle use, stridor	$\text{SpO}_2 < 90\\%$ on room air, cyanosis, audible stridor, RR > 32 or < 8 bpm	Acute respiratory distress, upper airway obstruction, severe asthma
3	Cardiac Warning Signs	Chest pain character, radiation, diaphoresis, palpitations, HR	Crushing retrosternal pain radiating to jaw/left arm, syncope with chest pressure, HR > 150 or < 40 bpm	Acute Coronary Syndrome (ACS), malignant ventricular arrhythmia
4	Neurological Warning Signs	FAST criteria, Glasgow Coma Scale (GCS), pupil response, seizures	Acute focal neurological deficit (facial droop, arm drift, slurred speech), GCS < 12, status epilepticus	Acute ischemic stroke, intracranial hemorrhage, acute meningitis
5	Blood Pressure Abnormalities	Systolic BP, Diastolic BP, pulse pressure, end-organ symptoms	Hypertensive crisis ($\text{SBP} > 180$ or $\text{DBP} > 120$ mmHg with headache/chest pain), profound shock ($\text{SBP} < 85$ mmHg)	Malignant hypertension, hypertensive encephalopathy, hypovolemic shock
6	Diabetes-Related Risks	Capillary Blood Glucose (CBG), Kussmaul breathing, fruity breath	Severe hypoglycemia ($\text{CBG} < 60$ mg/dL with altered consciousness) or $\text{CBG} > 400$ mg/dL with ketotic breathing	Hypoglycemic coma, Diabetic Ketoacidosis (DKA), Hyperosmolar Hyperglycemic State (HHS)
7	Gastrointestinal Issues	Abdominal pain quadrant, rebound tenderness, emesis, stool color	Massive hematemesis, involuntary abdominal guarding/rigidity, melena with hemodynamic instability	Perforated hollow viscus, active variceal upper GI hemorrhage
8	Injury & Trauma	Trauma mechanism, bleeding rate, deformity, airway patency	Pulsatile arterial hemorrhage, flail chest, suspected acute spinal cord injury, traumatic amputation	Exsanguinating hemorrhage, tension pneumothorax, major polytrauma

No.	Risk Domain	Required Inputs & Assessment Vectors	Tier-1 Red Flags (Hard Override to Critical)	Clinical Acuity Focus
9	Pregnancy-Related Risks	Gestational age (> 20 wks), pelvic pain, bleeding, headache	Third-trimester vaginal hemorrhage, severe epigastric pain with visual aura (pre-eclampsia), eclamptic convulsions	Placental abruption, severe pre-eclampsia, eclampsia
10	Allergic Reactions	Skin urticaria, mucosal swelling, voice change, wheeze	Angioedema of tongue/pharynx, acute stridor, wheezing combined with systemic hypotension	Anaphylactic shock, acute fatal airway closure
11	Dehydration & Fluid Balance	Skin turgor, mucous membrane status, urine output, cap refill	Anuria > 12 hours, severe skin tenting, cold extremities with capillary refill time > 4 seconds	Severe hypovolemia, pre-renal acute renal failure
12	Multi-System / Toxic Exposure	Substance ingested, puncture marks, salivation, pupil size	Organophosphate poisoning signs (SLUDGE), venomous bite with ptosis	Acute chemical toxicity, neurotoxic envenomation

6. Cross-Category Vital Signs Safety Baselines

Vital Sign Parameter	Severely Abnormal (Critical Trigger)	Moderately Abnormal (Warning)	Normal Adult Baseline
Heart Rate (HR)	< 40 bpm or > 150 bpm	40–49 bpm or 110–149 bpm	50–100 bpm
Respiratory Rate (RR)	< 8 bpm or > 32 bpm	22–32 bpm	12–20 bpm
Systolic BP (SBP)	< 85 mmHg or > 180 mmHg	85–89 mmHg or 140–179 mmHg	90–120 mmHg
Diastolic BP (DBP)	> 120 mmHg	90–119 mmHg	60–80 mmHg
Oxygen Saturation (SpO_2)	< 90% on room air	90%–93%	95%–100%
Body Temperature	< 35.0°C or > 40.0°C	35.0°–35.9°C or 38.3°–39.9°C	36.5°–37.5°C
Capillary Glucose (CBG)	< 60 mg/dL or > 400 mg/dL	60–69 mg/dL or 200–399 mg/dL	70–140 mg/dL (random)

7. Triage Acuity Stratification & Mathematical Formulation

- **Critical (Score 85–100) — Priority 1 (Color:** `#E53E3E`**):** Immediate life threat. Continuous monitoring. Physician evaluation required in 0 minutes (immediate bay allocation).
- **High (Score 60–84) — Priority 2 (Color:** `#ED8936`**):** Emergent condition. High deterioration probability. Physician assessment within 15 to 30 minutes.
- **Moderate (Score 30–59) — Priority 3 (Color:** `#ECC94B`**):** Urgent condition requiring diagnostic investigation. Evaluation within 60 minutes.
- **Low (Score 0–29) — Priority 4 (Color:** `#38A169`**):** Non-urgent ambulatory condition. Routine clinic assessment within 120 minutes.

Scoring Formula & Weights

Scoring Formula

```
Score = W_V · S_{\text{vitals}} + W_S · S_{\text{symptoms}} + W_D · S_{\text{demographics}} + S_{\text{delta}}
```

- $S_{\text{vitals}} \in [0, 100]$: Multi-parameter physiological deviation score based on weighted vital abnormalities.
- $S_{\text{symptoms}} \in [0, 100]$: Probability output from the gradient boosted symptom ensemble.
- $S_{\text{demographics}} \in [0, 100]$: Risk modifier (Age > 65, known immunosuppression, severe chronic comorbidity).
- $S_{\text{delta}} \in [0, 20]$: Physiological trajectory index measuring deterioration rate between consecutive vital captures.

• **Normalized Weights:** $W_V = 0.50$, $W_S = 0.35$, $W_D = 0.15$.

8. Explainable AI (XAI) & Missing-Data Protocols

• **Explainable AI:** Every risk score must be accompanied by an explainability payload indicating top contributing features computed via TreeSHAP with latency < 100 ms.

• **Missing Essential Vitals:** If SpO_2, SBP, or Heart Rate are unrecorded, confidence is flagged as `"LOW"`, uncertainty interval expands, and a mandatory clinical warning prompt is enforced in the UI.

• **Missing Non-Essential Data:** Demographics or symptom durations are imputed via population medians conditioned on patient age/gender strata; every imputed variable is logged in metadata.

9. Multilingual Engine & Voice Symptom Input

• **Bilingual Localization (English & Tamil):** Dynamic runtime switching between English (`en-US`) and Tamil (`ta-IN`). Key medical terms mapping:

– "Chest Pain" $\leftrightarrow$ "[illegible]"

– "Shortness of Breath" $\leftrightarrow$ "[illegible]"

– "Dizziness / Fainting" $\leftrightarrow$ "[illegible]"

– "Severe Abdominal Pain" $\leftrightarrow$ "[illegible]"

• **Voice Input Pipeline:** Uses browser-native Web Speech API (`webkitSpeechRecognition`) with server-side OpenAI Whisper container as an offline fallback.

10. Role-Based Access Control (RBAC) & Security Architecture

Role	Allowed API Endpoints & Actions	Protected Health Information (PHI) Access
Anonymous / Patient	Self-intake submission, retrieve self-triage status token.	Access limited to self-submitted intake form only.
Triage Nurse	Full vital intake, triage queue review, add clinical notes.	Access to active department patients (MRN, Name, Vitals, Triage Tier).
Attending Doctor	Comprehensive patient review, override triage tier, assign disposition.	Full access to clinical history, vitals trends, and override ledger.
System Administrator	Model versioning, system configuration, audit review.	Zero PHI visibility (all patient identifiers masked with SHA-256).

• **Data at Rest:** Encrypted using AES-256-GCM.

• **Data in Transit:** TLS 1.3 enforced.

• **Audit Logging:** Immutable log tracking user login, triage assessment, doctor override, and record viewing with payload hashes.

11. Relational Database Schema (PostgreSQL DDL)

```
-- Enable cryptographic extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- Table: patients (Encrypted PHI at Rest)
CREATE TABLE patients (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    mrn_hash VARCHAR(64) UNIQUE NOT NULL,
    encrypted_name BYTEA NOT NULL,
    date_of_birth DATE NOT NULL,
    gender VARCHAR(16) NOT NULL,
    primary_language VARCHAR(8) DEFAULT 'en',
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
```

```
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Table: vital_signs (Snapshot of clinical observations)
CREATE TABLE vital_signs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,
    heart_rate INT CHECK (heart_rate BETWEEN 20 AND 300),
    respiratory_rate INT CHECK (respiratory_rate BETWEEN 4 AND 80),
    systolic_bp INT CHECK (systolic_bp BETWEEN 40 AND 300),
    diastolic_bp INT CHECK (diastolic_bp BETWEEN 20 AND 200),
    oxygen_saturation NUMERIC(4,1) CHECK (oxygen_saturation BETWEEN 40.0 AND 100.0),
    temperature_celsius NUMERIC(4,2) CHECK (temperature_celsius BETWEEN 30.0 AND 45.0),
    capillary_blood_glucose NUMERIC(5,1),
    recorded_by UUID NOT NULL,
    recorded_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Table: triage_assessments (Core Decision-Support Output)
CREATE TABLE triage_assessments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,
    vital_id UUID REFERENCES vital_signs(id),
    chief_complaint TEXT NOT NULL,
    symptom_vector JSONB NOT NULL,
    calculated_risk_score INT NOT NULL CHECK (calculated_risk_score BETWEEN 0 AND 100),
    calculated_category VARCHAR(16) NOT NULL CHECK (calculated_category IN ('LOW', 'MODERATE', 'HIGH', 'CRITICAL')),
    hard_red_flag_triggered BOOLEAN NOT NULL DEFAULT FALSE,
    red_flag_details TEXT[],
    confidence_score NUMERIC(3,2) NOT NULL,
    explainability_payload JSONB NOT NULL,
    status VARCHAR(24) NOT NULL DEFAULT 'PENDING_REVIEW' CHECK (status IN ('PENDING_REVIEW', 'CONFIRMED', 'OVERRIDDEN',
'DISCHARGED')),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Table: triage_reviews (Doctor Review & Clinical Override Audit Ledger)
CREATE TABLE triage_reviews (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    assessment_id UUID NOT NULL REFERENCES triage_assessments(id) ON DELETE RESTRICT,
    doctor_id UUID NOT NULL,
    original_category VARCHAR(16) NOT NULL,
    final_category VARCHAR(16) NOT NULL,
    is_overridden BOOLEAN GENERATED ALWAYS AS (original_category != final_category) STORED,
    override_reason TEXT,
    clinical_notes TEXT,
    reviewed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Table: triage_audit_log (Immutable, Append-Only System Audit Journal)
CREATE TABLE triage_audit_log (
    id BIGSERIAL PRIMARY KEY,
    event_type VARCHAR(64) NOT NULL,
    user_id UUID,
    patient_id UUID,
    ip_address INET NOT NULL,
    user_agent TEXT,
    payload JSONB,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);
```

12. Standardized API Architecture & Contracts

- **Success Envelope:** `{"data": {}, "meta": {}, "error": null}`
- **Error Envelope:** `{"data": null, "meta": {}, "error": {"code": "CODE", "message": "Human message", "details": []}}`

Core Endpoints:

POST /api/v1/triage/assess — Execute Triage Risk Assessment.

GET /api/v1/doctor/queue?status=PENDING_REVIEW — Retrieve Doctor Triage Queue.

POST /api/v1/doctor/override — Physician Review & Override.

13. UI/UX Design System & Workstation Workflows

• **Semantic Acuity Colors:**

– **Critical (P1):** #E53E3E (Red)

– **High (P2):** #ED8936 (Orange)

– **Moderate (P3):** #ECC94B (Yellow)

– **Low (P4):** #38A169 (Green)

• **Accessibility Mandate:** WCAG AA contrast compliance (> 4.5:1), keyboard accessibility, and numerical labels required alongside color coding.

14. Testing Architecture & Validation Protocol

• **Deterministic Unit Tests:** 100% test branch coverage on the Clinical Red-Flag Safety Engine.

• **ML Regression & Drift Tests:** Evaluated against retrospective datasets.

• **Integration & API Contract Tests:** Automated validation matching OpenAPI 3.1 schemas.

• **End-to-End Simulation Tests:** Full intake-to-override simulated workflows.

15. Master Project Directory Structure

```
MediAI/
■■■ README.md, LICENSE, .gitignore, .env.example
■■■ docs/ (master/, architecture/, api/, database/)
■■■ app/ (entry/, layouts/, routes/, providers/)
■■■ ml/ (datasets/, models/, pipelines/, tests/)
■■■ shared/ (components/, hooks/, services/, utils/, types/, constants/, locales/, styles/)
■■■ modules/ (patient_intake/, triage_engine/, doctor_dashboard/, audit_compliance/)
■■■ migrations/
■■■ tests/ (unit/, integration/, api/, e2e/)
```

16. Definition of Done (DoD) & AI Coding Agent Protocol

• Acceptance criteria fully met, 0 regressions on safety tests, standardized JSON envelopes used, server-side RBAC applied, zero raw PHI exposed, WCAG AA compliance verified.

• **Mandatory AI Rules:** Read-First Mandate, Zero Fabricated Tests, No Unsanctioned Schema Edits, Boundary Respect, and Fail-Safe Integrity (never loosen red-flag rules).